



Golden Hospital Discharge Record



A Consolidated, Explainable Discharge-Level Dataset



Overview

This project implements a **Golden Hospital Discharge Record** by consolidating multiple hospital source systems into a single, reliable dataset with **one row per discharge (IPID)**.

Healthcare data typically exists across multiple transactional and clinical systems, often resulting in one-to-many relationships that complicate reporting and analysis. This solution addresses that challenge by creating a **clean, explainable, and reproducible discharge-level dataset** suitable for reporting, billing analysis, and downstream analytics.



Objective

The primary objectives of this project are:

-  Build a consolidated discharge-level dataset from multiple hospital systems
-  Resolve one-to-many relationships without duplicating discharge records
-  Preserve data integrity and analytical explainability
-  Validate correctness using explicit, auditable checks



Source Datasets

The following datasets were provided as part of the assessment:



1. discharge_master.xlsx

Primary system-of-record containing **one row per discharge**.

This dataset was treated as the **anchor table** for the golden record.

2. discharge_details.xlsx

Procedure-level billing data containing **multiple records per discharge (IPID)**.

3. clinical_findings.xlsx

Clinical notes and diagnosis information, also containing **multiple records per discharge**.

Data Model and Design Decisions

-  discharge_master was treated as the **anchor dataset**
-  Each row in the final output represents **exactly one discharge (IPID)**
-  One-to-many datasets were **aggregated prior to joining**
-  Clinical data was reduced to **medically relevant summaries**
-  No artificial imputation or assumptions were applied to missing data

This design ensures correctness, transparency, and reproducibility.

Transformations Performed

1. Derived Fields

Length of Stay (LOS) was calculated as:

 **DISCHARGED_DATE – ADMITTED_DATE (in days)**

This provides a standardized measure of inpatient duration for analysis.

2. Procedure Aggregation

From discharge_details.xlsx, the following discharge-level metrics were derived:

-  **Total Procedure Amount**
(*SUM of AMOUNT per IPID*)
-  **Procedure Count**
(*COUNT of procedures per IPID*)

These aggregations eliminate row duplication while preserving billing information.

3. Clinical Data Handling

From clinical_findings.xlsx:

-  Patient demographics (GENDER, AGE) were extracted
-  Final diagnosis was derived using clinical description filtering
-  The **earliest available final diagnosis** was retained per IPID
-  Clinical notes were concatenated into a summarized text field

This approach balances clinical relevance with analytical usability.

Join Strategy

-  All joins were performed **after aggregation**
-  Join key used: **IPID**
-  Join type: **LEFT JOIN**, preserving all discharges
-  One-to-one joins enforced to prevent row explosion

This guarantees that the golden dataset maintains one row per discharge.

Validation Checks

To ensure correctness and reproducibility, the following checks were explicitly performed:

-  Final row count matches discharge_master
-  Each IPID appears **exactly once**
-  Aggregated billing totals match raw transactional data

These checks confirm the integrity of the golden record

Output

The final outputs generated are:

-  outputs/golden_discharge_record.csv
-  outputs/golden_discharge_record.xlsx

Each row corresponds to **one complete discharge event**.

How to Run

Option 1: Notebook Execution

1. Open Notebooks/golden_discharge.ipynb
2.  Run all cells sequentially
3.  Output files generated in the outputs/ directory

Option 2: Script Execution

python Notebooks/golden_discharge.py

Tools and Environment

-  Python 3
-  pandas
-  openpyxl

Notes

-  No visualizations created; focus is on **data modeling and consolidation**
-  Design prioritizes **correctness, clarity, and interview explainability**
-  Solution is **database-agnostic** and suitable for SQL pipeline

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